

Lecture 10 - Direct Simulation Monte Carlo

Rarefied gas flow simulation

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1 Objectives

In this lecture, we will learn about rarefied gas flows and how to simulate them with the Direct Simulation Monte Carlo Method.

1. Flow regimes
2. Boltzmann equation, collision mechanics
3. DSMC

2 Flow regimes

Key parameter governing type of regime: Knudsen number.

$$\text{Kn} = \frac{\lambda}{L},$$

λ - mean free path, L - problem length scale.

- $\text{Kn} \ll 1$ - continuum regime
- $\text{Kn} \gg 1$ - free-molecular regime

$\lambda \propto 1/n$, n - number density.

2.1 Numerical methods for different flow regimes

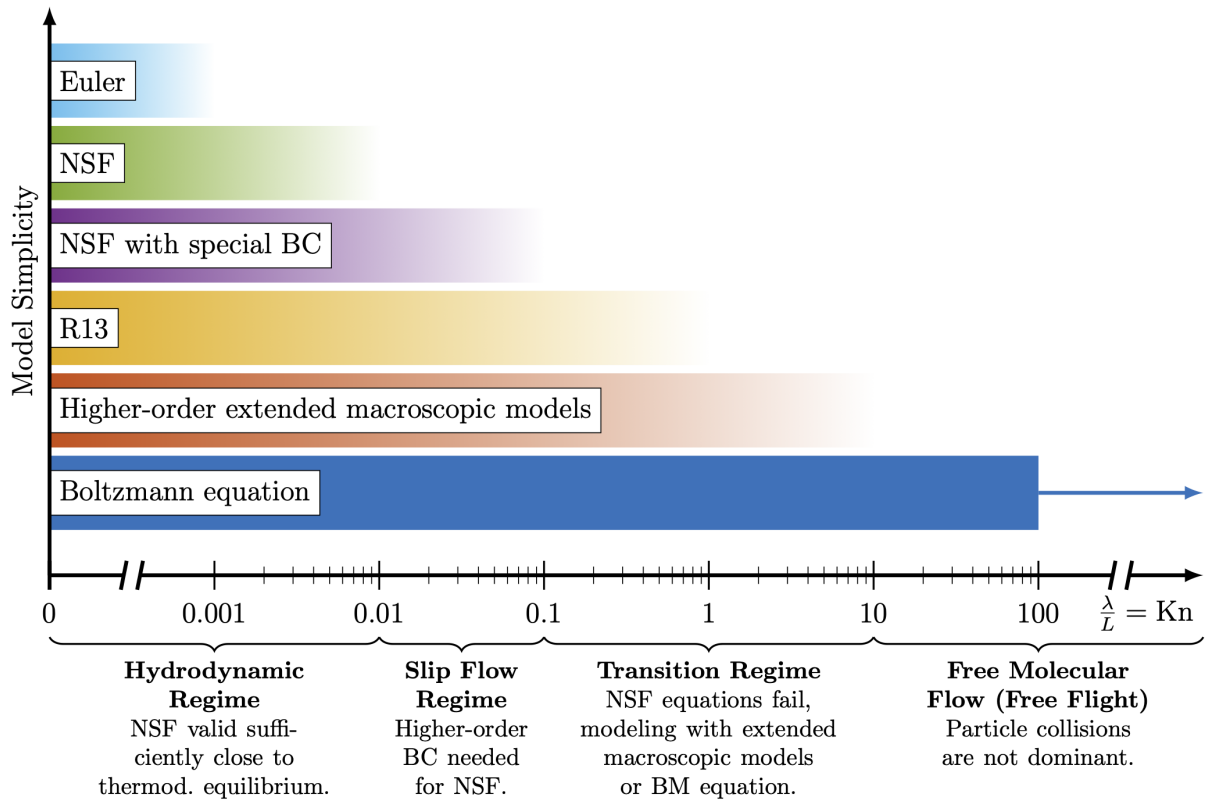


Figure 1: Applicability of models across Knudsen numbers

! Important

Problem length scale is usually not uniquely defined $\Rightarrow$ different regimes can coexist in the same problem.

2.2 Statistical flow description

What gives us more detail than the usual macroscopic quantities?

Distribution function:

$$f(\mathbf{x}, \mathbf{v}, t) \in \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}_+ \quad [\text{m}^{-3} \text{ m}^{-3}]$$

It is the expectation of the number of particles in the phase space volume $d\mathbf{x}d\mathbf{v}$ at time t .

2.3 Macroscopic quantities

Per definition:

$$n(\mathbf{x}, t) = \int f(\mathbf{x}, \mathbf{v}, t) d\mathbf{v},$$

$$n(\mathbf{x}, t) \mathbf{u}(\mathbf{x}, t) = \int \mathbf{v} f(\mathbf{x}, \mathbf{v}, t) d\mathbf{v},$$

$$\rho(\mathbf{x}, t) E(\mathbf{x}, t) = \int \frac{1}{2} m (\mathbf{v} - \mathbf{u})^2 f(\mathbf{x}, \mathbf{v}, t) d\mathbf{v}.$$

2.4 Absolute and peculiar velocities

Particle velocity $\mathbf{v}$ can be decomposed into

$$\mathbf{v} = \mathbf{u} + \mathbf{c},$$

where $\mathbf{c}$ is the “peculiar velocity” (velocity relative to the mean flow).

Non-central moments:

$$M_{ijk} = \int v_x^i v_y^j v_z^k f(\mathbf{x}, \mathbf{v}, t) d\mathbf{v},$$

central moments:

$$\hat{M}_{ijk} = \int c_x^i c_y^j c_z^k f(\mathbf{x}, \mathbf{v}, t) d\mathbf{v}.$$

3 Boltzmann equation

If we have an equation for f that is valid in **large range of regimes**, then we can solve it and compute ρ , $\mathbf{v}$, E , etc.

3.1 Kinetic equation

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla f = \mathcal{Q}(f).$$

Left-hand side is transport (hyperbolic), right-hand side is source term due to particle-particle interactions.

Question

What does the equation reduce to in the free-molecular regime?

3.2 Equilibrium distribution

For non-zero $\mathcal{Q}(f)$, the equilibrium distribution f_{eq} is a function that satisfies $\mathcal{Q}(f_{eq}) = 0$.

Important

Compared to free-molecular regime, this means that for $f \neq f_{eq}$ $\mathcal{Q}(f)$ is non-zero!

4 Binary collision mechanics

How do particles collide?

Binary collision of particles with masses m_1, m_2 leads to two conservation laws:

$$m_1 \mathbf{v}_1 + m_2 \mathbf{v}_2 = m_1 \mathbf{v}'_1 + m_2 \mathbf{v}'_2,$$

$$\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 = \frac{1}{2} m_1 (v'_1)^2 + \frac{1}{2} m_2 (v'_2)^2.$$

i Note

6 unknowns for the post-collision velocities, 4 scalar conservation equations.

4.1 Binary collision mechanics - center of mass velocity

Introduce center of mass velocity

$$\mathbf{v}_{cm} = \frac{m_1}{m_1 + m_2} \mathbf{v}_1 + \frac{m_2}{m_1 + m_2} \mathbf{v}_2.$$

Conservation of momentum implies

$$(m_1 + m_2) \mathbf{v}_{cm} = (m_1 + m_2) \mathbf{v}'_{cm}.$$

Center-of-mass velocity remains unchanged in collisions.

4.2 Binary collision mechanics - relative velocity

Relative velocity

$$\mathbf{g} = \mathbf{v}_1 - \mathbf{v}_2.$$

If we find the post-collision relative velocity $\mathbf{g}'$, then the post-collision velocities are

$$\mathbf{v}'_1 = \mathbf{v}_{cm} + \frac{m_2}{m_1 + m_2} \mathbf{g}', \quad \mathbf{v}'_2 = \mathbf{v}_{cm} - \frac{m_1}{m_1 + m_2} \mathbf{g}'.$$

4.3 Binary collision mechanics - conservation of energy

Energy conservation can be re-written as

$$(m_1 + m_2)v_{cm}^2 + m_r g^2 = (m_1 + m_2)v_{cm}^2 + m_r (g')^2,$$

where **collision-reduced mass** is given by

$$m_r = \frac{m_1 m_2}{m_1 + m_2}.$$

So magnitude of $\mathbf{g}$ remains unchanged - we only need to compute change of direction.

4.4 Binary collision mechanics - scattering cross-section

$$\chi = \chi(b, d),$$

differential scattering cross-section relates effective target area to outgoing solid angle:

$$\sigma_d := \frac{b}{\sin \chi} \left| \frac{db}{d\chi} \right|$$

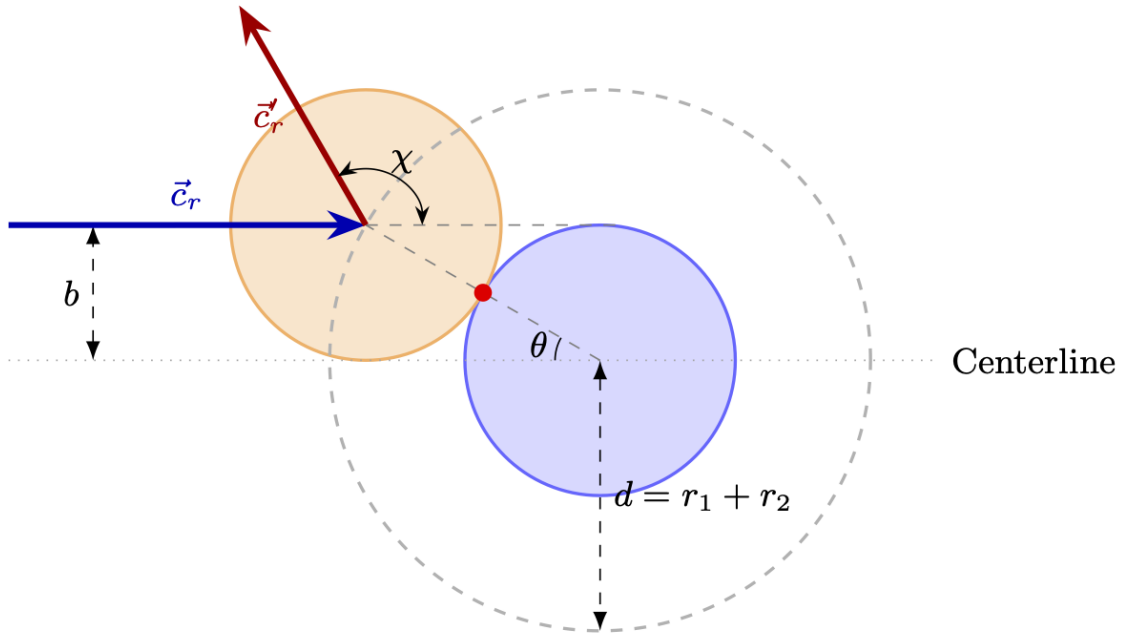


Figure 2: Binary collision mechanics

4.5 Total collision cross-section

Total collision cross-section:

$$\sigma_{tot} = 2\pi \int_0^{2\pi} \sigma_d \sin(\chi) d\chi.$$

Describes effective “size” of particles.

Common models:

- Hard spheres: $\sigma_{tot} = \pi d_{12}^2$, $d_{12} = (d_1 + d_2)/2$
- Variable hard spheres (VHS): $d(g) = d_{ref}(g_{ref}/g)^\nu$, $\chi = 2 \cos^{-1}(b/d)$
- Variable soft spheres: VHS + anisotropic scattering, $\chi = 2 \cos^{-1}((b/d)^{1/\alpha})$

Collision model defines transport properties (viscosity, thermal conductivity, etc.). Will omit subscript *tot* from now on.

4.6 Boltzmann collision operator

The Boltzmann collision operator describes the change in f due to collisions, and consists of a source $(\mathbf{v}', \mathbf{v}'_2) \rightarrow (\mathbf{v}, \mathbf{v}_2)$ and sink $(\mathbf{v}, \mathbf{v}_2) \rightarrow (\mathbf{v}', \mathbf{v}'_2)$ terms:

$$Q(f(\mathbf{v})) = \int (f(\mathbf{v}')f(\mathbf{v}'_2) - f(\mathbf{v})f(\mathbf{v}_2)) g \sigma_d d^2\Omega d\mathbf{v}_2$$

! Important

$\mathbf{v}'$ and $\mathbf{v}'_2$ are not arbitrary velocities! They depend on $\mathbf{v}$ and $\mathbf{v}_2$ via collision scattering law defined by σ_d .

4.7 Equilibrium distribution

For the Boltzmann collision operator, the equilibrium distribution is given by

$$f_{eq} = n \left(\frac{m}{2\pi kT} \right)^{3/2} \exp \left(-\frac{m(\mathbf{v} - \mathbf{u})^2}{2kT} \right)$$

i Note

1. $n = n(\mathbf{x})$, $\mathbf{u} = \mathbf{u}(\mathbf{x})$, $T = T(\mathbf{x})$
2. Oftentimes $\mathbf{v} - \mathbf{u}$ directly written as $\mathbf{c}$

4.8 BGK equation

A simplified model for the collision operator is the BGK (Bhatnagar-Gross-Krook) model:

$$Q(f) = \frac{f_{eq} - f}{\tau},$$

τ is the relaxation time.

Model cannot capture viscosity and thermal conductivity simultaneously ($Pr \equiv 1$) $\Rightarrow$ extensions include the Ellipsoidal-Statistical (ES-BGK) and Shakhov models.

4.9 Moment equations

Multiplying the Boltzmann equation by $(v_x)^i (v_y)^j (v_z)^k$, we obtain an equation for M_{ijk} :

$$\frac{\partial M_{ijk}}{\partial t} + \nabla \cdot (M_{i+1,jk}, M_{i,j+1,k}, M_{i,j,k+1}) = \int (v_x)^i (v_y)^j (v_z)^k Q(f) d\mathbf{v}.$$

💡 Question

What's missing here?

i Infinite moment hierarchy

The moment equations are coupled to each other, forming an infinite hierarchy - we need some assumption/model to compute flux based on known moments.

4.10 Euler equations, Navier-Stokes

Euler equations: 0, 1st, and 2-nd order (summed) moment equations + **assumption** that $f \equiv f_{eq}$. Can show that there is no heat transport, no viscosity, diagonal pressure tensor with $p = nkT$.

Navier-Stokes equations (Chapman-Enskog procedure): also 5 transport equations + **assumption** that $f \equiv f_{eq}(1 + \phi)$, $\phi = \phi(\nabla\rho, \nabla\mathbf{v}, \nabla T)$ is a small correction of order $\mathcal{O}(\text{Kn})$.

4.11 Multi-species inelastic collision integrals

In case of internal energy transitions and binary chemical reactions (2 reagents, 2 products):

$$Q_{ci}^{2\leftrightarrow 2} = \sum_{dc'd'} \sum_{ki'k'} \int \left(f_{c'i'} f_{d'k'} \frac{s_i^c s_k^d}{s_{i'}^c s_{k'}^d} \left(\frac{m_c m_d}{m_{c'} m_{d'}} \right)^3 - f_{ci} f_{dk} \right) g \sigma_{cd,ik}^{c'd',i'k'} d^2\Omega d\mathbf{v}_d.$$

5 DSMC

Boltzmann equation is **extremely** expensive computationally.

Semi-discretization in time requires discretization of 6-dimensional VDF.

A DVM method: 20^3 nodal values in velocity space at each fixed $\mathbf{x}$: $8000 \times N_{cells}$ values to store and update.

A stochastic particle method would trade memory usage/computational speed for noise.

5.1 Key ideas of DSMC

- Whole flow is represented by particles. Unlike SPH, there is no smoothing: properties are computed in a cell based on the particles' properties in the cell
- Each particle represents $F_{num} \equiv const$ physical particles
- Particles move in time using explicit time-stepping (forward Euler), colliding with walls in the process
- Particle collisions are modelled **stochastically** between particles in a **collision cell**, locations of particles within cell are irrelevant
- In steady state, ensemble averaging $\Leftrightarrow$ time-averaging

5.2 Flow initialization

Given

- number density n
- mean flow velocity $\mathbf{u}$
- temperature T
- cell volume V
- F_{num}

one computes

- number of particles to sample: $N_{p,cell} = nV/F_{num}$
- samples v_x, v_y, v_z components from Gaussian distributions with mean 0 and standard deviation $\sqrt{k_B T/m}$ (Maxwellian distribution)
- add mean flow velocity to the sampled velocities
- particle positions are sampled uniformly in the cell

5.3 Particle-surface interactions

Simplest models (assuming wall normal to x direction)

- specular reflection: $v'_x = -v_x$
- fully diffuse wall: $\mathbf{v}'$ is sampled from Maxwellian distribution with wall temperature and velocity
- Maxwell's model: $(1 - \alpha)R_{spec} + \alpha R_{diffuse}$, α - accommodation coefficient
- other more complex models: Cercignani-Lampis (independent tangential and normal accommodation coefficient)

5.4 Particle-particle collisions

Assume we have chosen two particles to collide (how? discussed further below). DSMC follows the procedure:

- compute $\mathbf{v}_{cm}, \mathbf{g}$
- for isotropic scattering, sample point $\mathbf{w}'$ on unit sphere S^2
- compute $\mathbf{g}' = |\mathbf{g}|\mathbf{w}'$
- compute post-collision particle velocities via $\mathbf{v}_{cm}, \mathbf{g}'$

In case of anisotropic scattering, change in $|\mathbf{g}'|$ (inelastic processes): procedure similar, but more involved (see Boyd, Schwartzentruber.)

5.5 Collision selection

Given a set of particles in a cell, how do we choose which two particles to collide? The simplest scheme (“ N^2 scheme”) iterates over all pairs of particles:

1. For pair i, j compute $p_{ij} = \sigma(g)gF_{num}\Delta t/(2V)$ - relative collision volume
2. If $\mathcal{R} < p_{ij}$, $\mathcal{R} \in \mathcal{U}[0, 1]$, perform collision

Of course, very expensive

5.6 Time counter scheme

Average time between collisions:

$$\tau = \frac{1}{Z}, \quad Z = \frac{N_p(N_p - 1)\langle\sigma g\rangle F_{num}}{2V}.$$

We don't know time average $\langle\sigma g\rangle$ of σg in advance, therefore:

1. Choose random pair i, j
2. Compute $\delta t = (2V)/(N_p(N_p - 1)\sigma(g)gF_{num})$
3. Compute collision
4. Update time counter $t_c += \delta t$ ($t_c = 0$ at start of collision step)
5. If $t_c > \Delta t$, stop

i Note

Scheme performs poorly if collision with small $\sigma(g)g$ is selected - probability of collision is small, but time counter immediately exceeds Δt , leading to wrong collision rates.

5.7 NTC scheme

Define

$$N_{\text{coll}} = \frac{N_p(N_p - 1)(\sigma g)_{\text{max}} F_{\text{num}} \Delta t}{V}.$$

- sample N_{coll} pairs
- compute g, σ for each pair
- if $\mathcal{R} > \sigma g / (\sigma g)_{\text{max}}$, $\mathcal{R} \in \mathcal{U}[0, 1]$ reject pair
- otherwise, perform collision

The scheme has linear cost: increasing resolution by factor of X increases N_p by factor of X , decreases F_{num} by factor of $X \implies N_{\text{coll}}$ is increased by a factor of X .

5.8 DSMC convergence

- Second-order in space and time
- But if we use a nearest-neighbour search algorithm, the scheme becomes first-order in time + we **always** need to track 3-dimensional position
- Signal-to-noise ratio is proportional to $1/\sqrt{N_p}$
- Signal-to-noise ratio is proportional to $1/\sqrt{Ma}$

5.9 DSMC convergence

In fact, error in space and time discretization is directly related to whether mean free path λ and mean collision time τ are resolved. Analysis of viscosity (hard sphere model):

$$\mu = \frac{5}{16d^2} \sqrt{\frac{mkT}{\pi}} \left(1 + \frac{16}{45\pi} \left(\frac{\Delta x}{\lambda} \right)^2 \right).$$

Similar error estimate in time:

$$\epsilon \propto \left(\frac{\Delta t}{\tau} \right)^2.$$

5.10 DSMC convergence

DSMC discretization requires:

- $\Delta x < \lambda$ (don't forget that $\lambda = \lambda(\mathbf{x})$)
- $\Delta t < \tau$ (don't forget that $\tau = \tau(\mathbf{x})$)
- $\Delta t v_{\text{max}} < \Delta x$ (CFL-like condition)

5.11 Variable-weight DSMC

Question

What's a drawback of fixed-weight DSMC ($F_{num} \equiv const$)?

Note

Fixed ratio of real-to-computational particles F_{num} leads to poor statistics for trace populations (density gradients, trace chemical species, high-velocity particles, etc.).

Using per-species/per-particle weights requires

1. New collision routines
2. Merging of particles to keep particle count within bounds

6 Particle-in-Cell

Particle-based method for the Vlasov-Poisson/Vlasov-Maxwell equation:

$$\frac{\partial f_c}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{x}} f_c + \frac{q_c}{m_c} (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \nabla_{\mathbf{v}} f_c = 0.$$

$$\nabla \cdot \mathbf{E} = \frac{\rho^c}{\varepsilon_0}, \quad \nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

6.1 Electrostatic equation:

$$\Delta \Phi = -\frac{\rho^c}{\varepsilon_0}, \quad \rho^c = \sum_c q_c n_c.$$

n_c computed via f_c , q_c is charge of particles of species c .

Solve Poisson equation for Φ on grid via method of choice (FD, DG, FEM, spectral, etc.).

To compute ρ^c on grid, we interpolate charge from particles onto grid. We then solve Poisson equation, compute electric field, interpolate field back onto particles.

7 Open-source DSMC codes

- [SPARTA](#)
- [dscmFoam+](#)
- [PANTERA](#): plasmas with DSMC collisions
- [PICLas](#): plasmas with DSMC collisions
- [Merzbild.jl](#): experimental variable-weight DSMC code
- [ParticLas.jl](#): DSMC-BGK code with on-the-fly visualization

8 Questions

- What is the Knudsen number and what are the different possible flow regimes? How does one define a statistical gas description and use it to compute macroscopic quantities?
- How do binary collision mechanics work?
- What are the main assumptions of the DSMC method? How do different collision routines function, and what are their advantages/disadvantages?

9 References

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